

FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 – 3.2 ROTATION: FINAL EXPLANATION	
Student Assessment Document	
Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the unit. Think back to the Ngong Hills Wind Farm video and everything you found out in Lessons 1 to 10. Answer each section in the space under it. Write in full sentences, use the correct rotation words (centre of rotation, angle of rotation, direction, object, image, order of rotational symmetry, congruent) and show any working or constructions you use. Always connect your answers to the Ngong Hills turbine. Positive angles mean anti-clockwise and negative angles mean clockwise.

Section 1: What is a rotation and what stays the same?

A friend watches the Ngong Hills turbine blades spinning and says, 'The blades move so much, so how can the turbine always look the same?'

- (a) Name the three things you need to state to describe a rotation completely, and say which part of the turbine is the centre of rotation.
- (b) State TWO properties of rotation that stay the same when the blades turn, and say what changes.
- (c) Explain how a positive angle and a negative angle differ.

Write your answer here:

Section 2: Predicting where a point lands

Treat the hub of the turbine as the origin $O(0,0)$. A blade tip is at $P(4,2)$.

- (a) Find the coordinates of the image of P after a rotation of 90° anti-clockwise about O .
- (b) Find the image of P after a rotation of 180° about O .
- (c) Find the image of P after a rotation of 270° anti-clockwise about O .
- (d) What happens to the blade tip after four successive 90° anti-clockwise rotations? Explain how this helps to answer the driving question.

Write your answer here:

Section 3: Why three blades look identical

- (a) The three blades are equally spaced around the hub. What is the angle between neighbouring blades?
- (b) What is the order of rotational symmetry of the three-bladed turbine? List the angles within one full turn at which the assembly maps onto itself.
- (c) Write the formula connecting the minimum angle of rotation and the order, and use it for the turbine.
- (d) Explain why the turbine looks identical after each of these turns.

Write your answer here:

Section 4: Congruence and finding a hidden centre

- (a) Blade 2 is the image of Blade 1 under a rotation about the hub. State the angle and direction of this rotation, and explain why this proves the three blades are congruent. Use the word isometry.**
- (b) You are shown only a blade in two positions (object and image) and the hub is hidden. Describe step by step how you would find the centre of rotation and then the angle of rotation.**
- (c) Explain why your method for finding the centre works.**

Write your answer here:

Section 5: Final Explanation

Answer the driving question in your own words: Why do the blades of the Ngong Hills wind turbine always look the same no matter how far they rotate, and how can we use the mathematics of rotation to predict exactly where any point on the blade will land? Use the terms centre of rotation, 120° , order of rotational symmetry, congruent/isometry, and positive/negative angle. Include evidence from the unit.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Rotation vocabulary and properties	Correctly names centre, angle and direction, states distance invariance and congruence, and uses positive/negative correctly throughout.	States most properties and terms correctly with minor slips in wording or sign convention.	Confuses terms or properties, or omits direction or what stays the same.
Coordinate calculations	Applies all rotation rules correctly to get $(-2,4)$, $(-4,-2)$, $(2,-4)$ and explains the return to $(4,2)$ after four turns.	Gets most images correct with a minor error or incomplete explanation of the full turn.	Uses wrong rules or cannot find image coordinates.
Rotational symmetry reasoning	Gives 120° spacing, order 3, angles 120° , 240° , 360° , uses $360^\circ \div \text{order}$, and links clearly to why the turbine looks identical.	Gives order and angle correctly but the link to the phenomenon is thin.	Cannot determine order or angle or does not link to the turbine.
Congruence and finding the centre	Uses isometry to justify congruence and describes the perpendicular bisector method fully with a valid reason it works.	Describes the method with most steps but the reason is incomplete.	Method is missing or incorrect, or congruence is asserted without justification.
Final Explanation quality	Coherent explanation using all required terms with accurate evidence from the unit, answering both parts of the driving question.	Answers the driving question with most terms and some evidence.	Answer is vague, lacks key terms or evidence, or answers only part of the question.

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